Business Questions, Diagnostics & Insight Validation

Telecom Customer Churn Analysis

This stage moves from exploration into answers: it addresses the required and self-generated business questions about churn drivers, builds diagnostic customer segments, quantifies the business impact of churn in revenue terms, and validates the resulting insights before they are handed off as recommendations.

Setup

Reuses the Stage 05 cleaned dataset, including engineered fields (churn_flag, tenure_group, monthly_charge_band), and the same visualization conventions established during EDA. All outputs from this notebook are exported to a single OUTPUT_PATH folder at the end. A sanity check confirms the dataset still carries zero missing values before analysis begins.

Section 7 — Required Business Questions

Answers the core questions the analysis was commissioned to address: which contract, payment, and service attributes are associated with churn, and which customer combinations concentrate the highest risk.

7.1 — What is the churn rate by contract type?

Groups by contract, computing customer count, churned count, and churn rate; visualized as a bar chart.

7.2 — What is the churn rate by payment method?

Same approach by payment method, visualized as a horizontal bar chart sorted by rate.

7.3 — What is the churn rate by internet service?

Same approach by internet service type.

7.4 — How do tenure and monthly charges differ between churned and retained customers?

Compares mean/median tenure and monthly charges by churn status, with box plots for each.

7.5 — Which customers combine month-to-month contracts with high monthly charges?

Flags customers in month-to-month contracts with charges at or above the 75th percentile as a first compounded-risk segment.

7.6 — Which subscribed services are most associated with churn?

Loops over six optional service columns to compute churn rate by category for each.

7.7 — At which tenure stage is churn highest?

Churn rate by tenure_group, visualized as a bar chart.

7.8 — Does contract type change the relationship between tenure and churn?

Cross-tabulates contract × tenure_group churn rate and visualizes it as a heatmap.

7.9 — Which contract × payment method combinations are highest risk?

Same cross-tab/heatmap approach for contract × payment method.

7.10 — Does the number of subscribed services relate to churn?

Builds a service_count feature (count of 'Yes' across the six service columns) and computes churn rate by count, visualized as a bar chart.

7.11 — Which demographic groups have higher churn?

Churn rate by senior-citizen status, partner status, and dependents, combined into one comparison table.

7.12 — Who are the highest-risk customer segments?

Groups by contract × tenure_group × internet_service, filters to segments with at least 50 customers to avoid noise, and ranks by churn rate.

7.13 — High-value + high-risk customers

Identifies customers in the top quartile of monthly charges who also churned — the clearest immediate revenue loss.

7.14 — How much monthly revenue is at risk overall?

Sums monthly_charges across all churned customers.

7.15 — Which high-risk segments carry the greatest revenue exposure?

Multiplies each Section 7.12 segment's customer count by its average monthly charge to rank by revenue exposure.

7.16 — Final executive risk ranking

A compact, presentation-ready top-10 ranking combining sample size, churn rate, and revenue exposure.

Section 8 — Self-Generated Questions

Motivated by patterns from the Stage 06 EDA: how churn evolves across the customer lifecycle, whether higher-paying customers are more or less loyal, and where revenue risk concentrates once tenure and internet service combine with pricing.

8.1 — Does churn risk change significantly across the customer lifecycle?

Churn rate and average monthly charge by tenure_group, visualized as a line chart.

8.2 — Does the impact of monthly charges depend on contract type?

Groups by contract × monthly_charge_band, visualized as a grouped bar chart.

8.3 — Which customers represent the greatest revenue risk?

Groups by contract × internet_service, computing monthly revenue and estimated revenue at risk.

8.4 — Are high-value customers actually more loyal?

Tests the assumption directly by comparing churn rate across monthly-charge tiers.

Answer (8.4): higher monthly charges are not, by themselves, a sign of loyalty — the high-charge tier does not show a lower churn rate than lower tiers, consistent with contract type and service quality mattering more than price alone.

Section 9 — Diagnostic / Segmentation Analysis

Moves from single-variable questions to deeper diagnostics: how service adoption interacts with loyalty, which service combinations compound risk, which segments to prioritize, and whether a composite risk score can summarize churn exposure in one number.

9.1 — Does having more services make customers more loyal?

Churn rate by service_count, visualized as a line chart.

9.2 — Which service combinations are associated with the highest churn?

Cross-tabs online_security × tech_support churn rate as a heatmap.

9.3 — Which customer segment should the company prioritize first?

Groups by contract × tenure_group (min. 50 customers), ranked by estimated revenue at risk.

9.4 — Can we create a practical customer risk profile?

Combines five binary risk flags — month-to-month contract, tenure ≤ 12 months, top-quartile monthly charge, no tech support, no online security — into a composite risk_score (0–5) and checks whether churn rate rises with the score.

Section 10 — Business Impact & Prioritization

Translates the diagnostic segments into dollar terms — how much monthly revenue churn represents, which contract and service lines are most exposed, and where high-value customers are being lost — so leadership can prioritize by financial impact rather than churn rate alone.

10.1 — Overall business baseline

Total customers, churned customers, overall churn rate, total monthly revenue, and monthly revenue represented by churned customers.

10.2 — Revenue exposure

Summarizes total monthly revenue, revenue from churned customers, and revenue exposure as a percentage.

10.3 — Revenue exposure by contract type

Monthly revenue and churned revenue by contract, with a revenue exposure rate, visualized as a horizontal bar chart.

10.4 — Revenue exposure by internet service

Same revenue-exposure calculation broken out by internet service type.

10.5 — High-value customers at risk

Revenue and customer counts for the top-quartile monthly-charge group, split by churn status.

10.6 — Revenue risk by customer segment

Re-uses the Section 7.12 contract × tenure_group × internet_service segmentation, now ranked by actual dollar exposure (churned revenue) rather than churn rate alone.

Section 11 — Insight Validation

Before the segments and rankings above are used to prioritize action, this section checks whether they hold up: are the differences statistically significant, are the segments backed by enough customers to trust, and does the composite risk score behave as intended?

11.1 — Are the observed churn differences statistically significant?

Runs a chi-square test of independence between churn and each of contract, payment method, internet service, and tenure_group, flagging significance at $p < 0.05$.

11.2 — Are the high-risk segments backed by a reliable sample size?

Flags which Section 7.12 segments meet a minimum threshold of 50 customers, since a striking churn rate on a tiny sample isn't a dependable target.

11.3 — Does the composite risk score behave as intended?

Checks whether churn rate rises monotonically as risk_score increases, confirming the score isn't skewed by a handful of extreme cases.

Conclusion: the segments and rankings from Sections 7–10 are statistically supported, backed by adequate sample sizes, and the composite risk score behaves consistently — ready to be used for prioritization.

Export

19 analysis outputs (churn-by-contract, churn-by-payment-method, churn-by-internet-service, churn-by-tenure-group, contract/payment risk, service and demographic churn analyses, segment analysis, high-value/high-risk customer lists, executive risk segments, lifecycle and charge/contract analyses, segment revenue, customer value tiers, service-loyalty analysis, security/support risk, priority segments, and risk-score analysis) are written as individual CSV files to the OUTPUT_PATH folder for downstream reporting and dashboarding.

These validated segments, revenue exposures, and the composite risk score are the inputs to the final recommendations / reporting stage.